

AN1829 - Understanding System Intrinsic Noise: Enabling Longer Reach and More Nodes on 10BASE-T1S Systems



IEEE Std 802.3cg™-2019 defines a 10 Mbit/sec Ethernet PHY that can support up to at least 8 nodes at a total bus length of up to at least 25 meters. There are many potential applications for 10BASE-T1S networks which require more than 8 nodes, or a total mixing segment length greater than 25 meters, or both. This can be achieved using Microchip LAN8670/1/2 and/or LAN8650/1. When used with an appropriately designed mixing segment, the Microchip devices can support up to 50 nodes and 50 meters with a significant noise margin, or 2 nodes at up to 100 meters in an environment with lower noise levels.



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1. Introduction

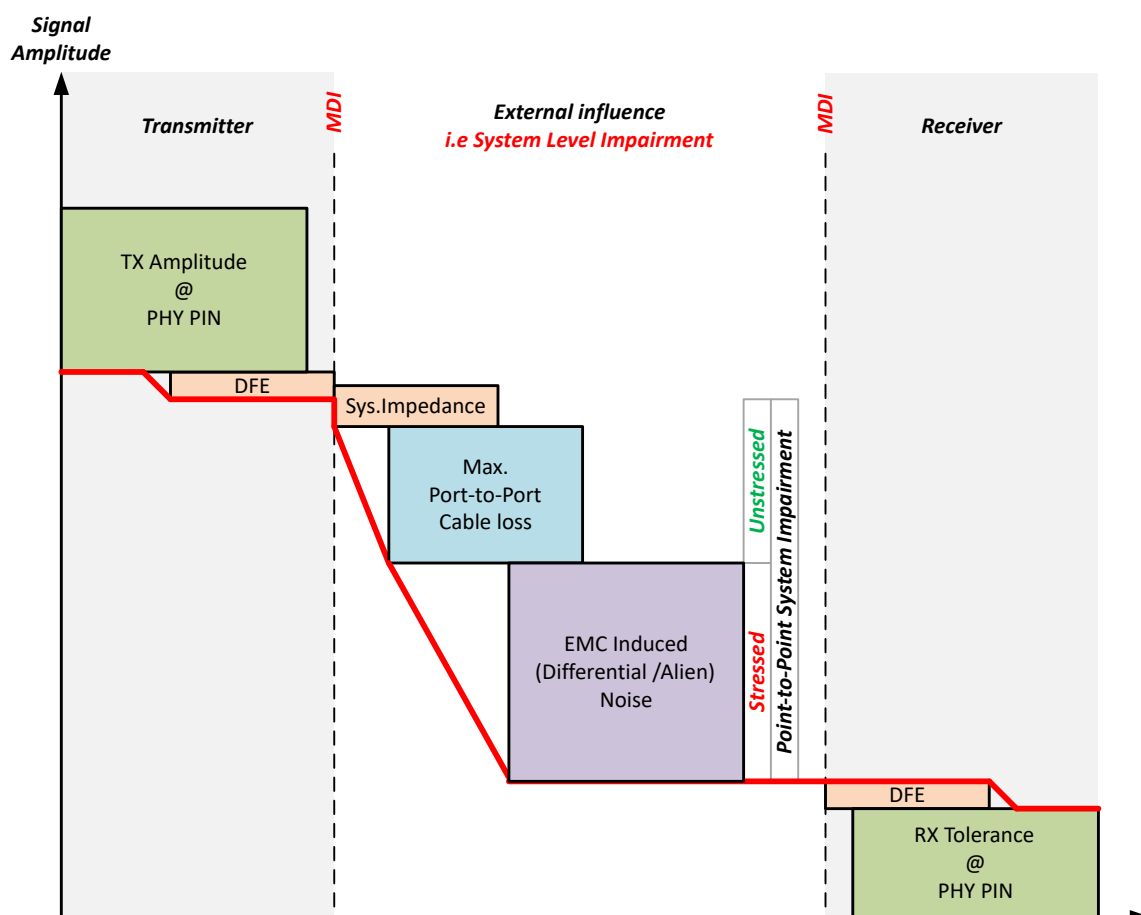
IEEE Std 802.3cg™-2019¹ defines a 10 Mbit/sec Ethernet PHY that can support up to at least 8 nodes at a total bus length of up to at least 25 meters. The PHY in the Microchip LAN8670/1/2² and LAN8650/1³ are designed to the IEEE standard and can certainly support 8 nodes at 25 meters. When used with an appropriate system of cable and connectors, the products can support many more nodes and much longer reach.

The standard does not specify the exact cable, although it does specify cable characteristics for a specific cable length. It also does not specify connectors, but these, as well as the network topology can determine whether the system can work reliably. This application note will show how the cable, total length, and bus topology determine the inherent noise of the mixing segment. Inherent noise reduces the noise margin of the system and determines whether a system can operate reliably in the appropriate noise environment: inherent noise determines how many nodes can be supported and how long the mixing segment can reach. This application note will show that with good cable, it is possible to design mixing segments to support well beyond 8 nodes and 25 meters.

2. Noise Budget

In any communications system, the signal at the receiver does not exactly match the signal at the transmitter. The sources of the difference in a point-to-point system are shown in Figure 2-1. The PHY interfaces to the cable via a Discrete Front End (DFE), such as the Bus Interface Network (BIN)⁴ recommended for use with the Microchip PHYs. The cable interface is named the Media Dependent Interface (MDI) in the IEEE specification and the standard defines the characteristics at exactly this point. The specification defines the media as a balanced pair and specifies some characteristics of the media, including insertion loss. Insertion loss is a measurement of the impedance mismatch between the MDI and the cable. This system impedance mismatch will cause reflections and therefore signal distortion. The attenuation over the cable length between any two ports will cause additional signal loss. These all combine to distort the signal, even before the addition of noise. A system that is built on better quality components and cabling will result in better signal quality by minimizing reflections and attenuation. This remaining difference between unstressed signal and the tolerance through the DFE of the receiver is the amount of noise that the system can tolerate. Noise can be from electromagnetic interference as well as from power supply and other noise sources in the receiver.

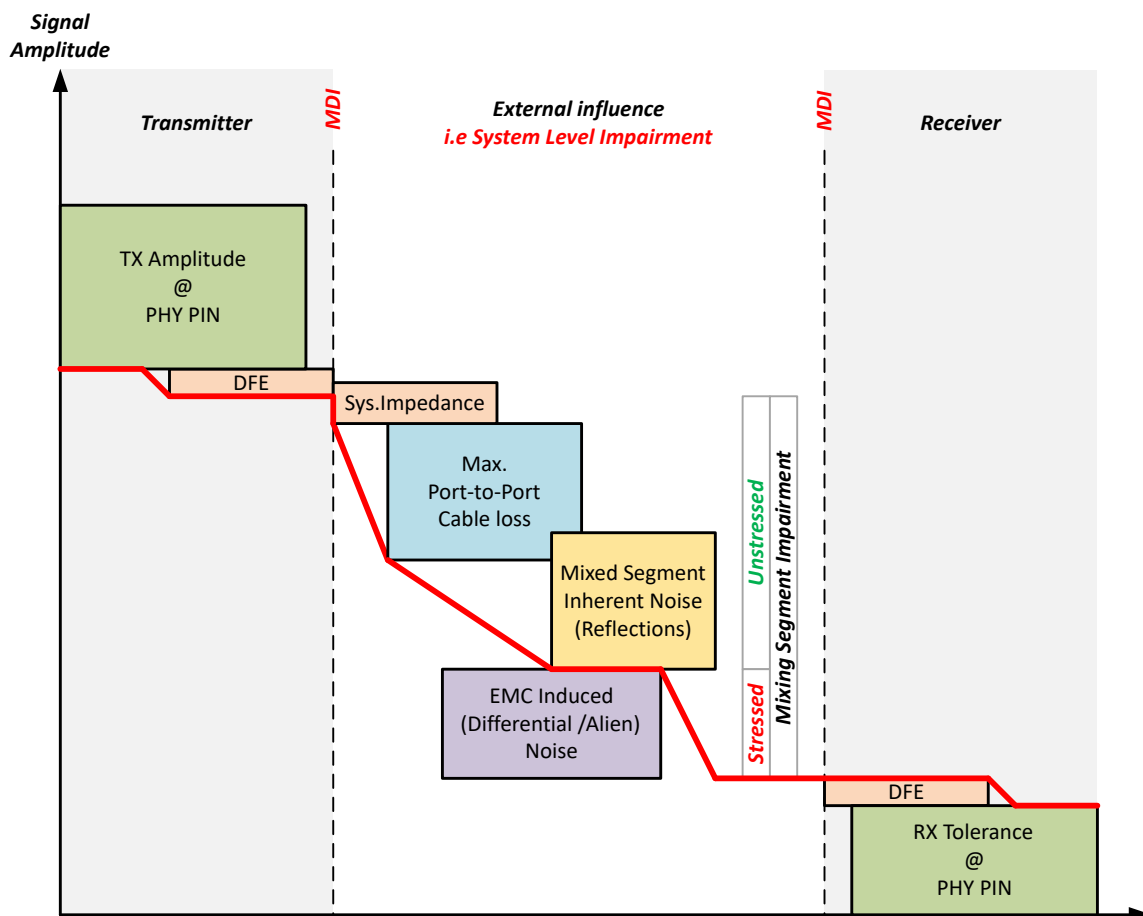
Figure 2-1. Noise Budget - Point to Point



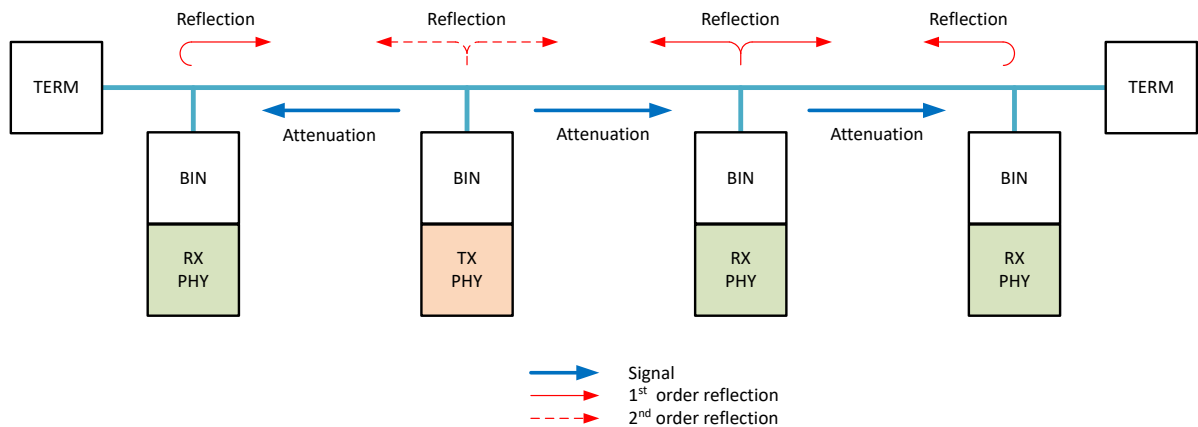
3. Inherent Noise in Mixing Segments

A 10BASE-T1S mixing segment normally contains more than 2 nodes and may include stubs; this results in more discontinuities than a point-to-point connection. Each discontinuity will contribute additional reflections and each segment will further attenuate the signal. This distortion of the signal caused by the mixing segment, is present even in the unstressed system, and will reduce the noise margin compared to a point-to-point network, as shown by the additional block in Figure 3-1.

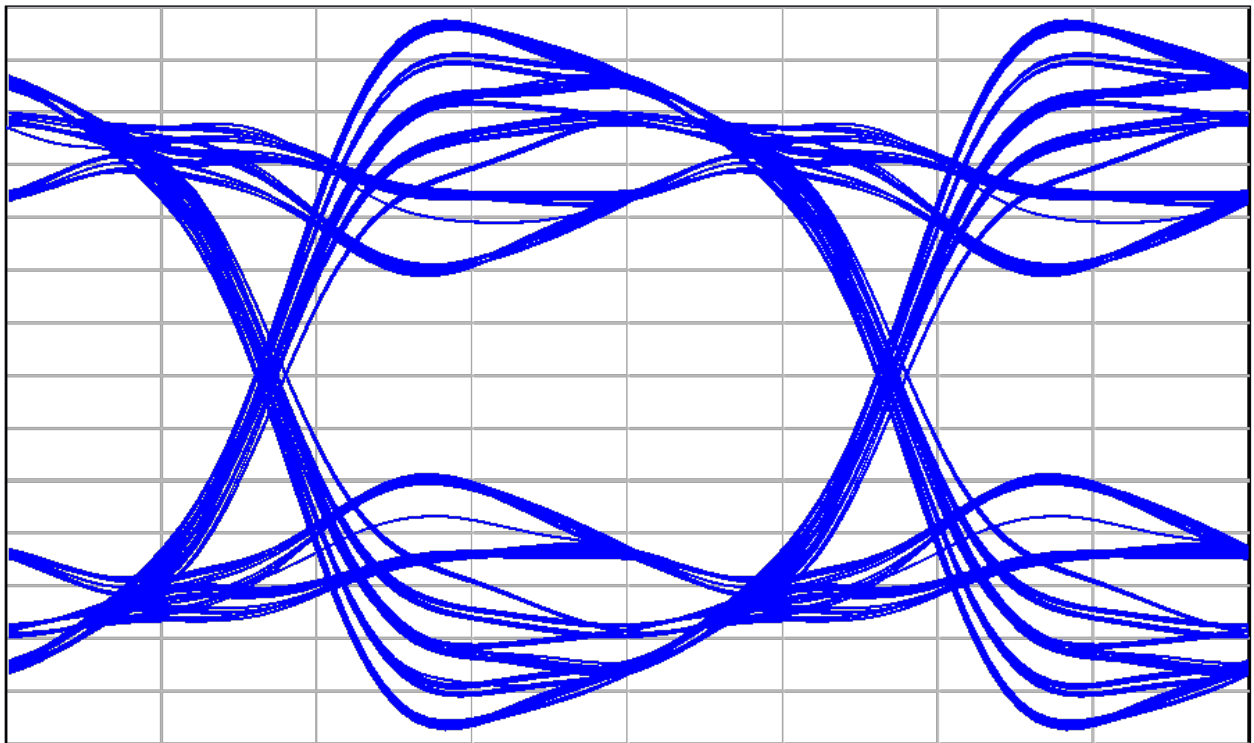
Figure 3-1. Noise Budget - Multidrop Mixing Segment



In a multi-drop system, in addition to the components and cabling, signal quality is also affected by the topology of the system. Each node, whether the node is the intended receiver or not, introduces a slight impedance mismatch, and therefore contributes a reflection of every signal that arrives. As shown in Figure 3-2, reflections cause their own reflections. Attenuation reduces the amplitude of signals. For a received signal, every node sees a different combination of reflections, and the combination depends on which node is transmitting. A topology in which many reflections occur very close to each other can result in a system in which some nodes never receive a good signal even though other nodes may work correctly.

Figure 3-2. Reflections in a Mixing Segment

The signal degradation that results from the combination of node hardware, cable characteristics and topology itself can be called *Inherent Noise*. For any pair of nodes, inherent noise will result in a less than ideal eye diagram. With a poor choice of cable and/or topology, inherent noise can be large enough to close the eye completely so that a receiver cannot recover the signal. Unlike other noise sources, inherent noise is fixed, and it is possible to use simulation or lab measurements to determine if a topology results in poor signal quality. An example of the eye pattern measurement showing the effects of inherent noise on a system with several nodes is shown in [Figure 3-3](#)

Figure 3-3. Eye Pattern Showing Effects of Inherent Noise

4. Cable Characteristics

IEEE Std 802.3cg™-2019 standard specifies a balanced pair of conductors. It does not require twisting of the conductors, nor does it require shielding. Instead, it defines the characteristics for a mixing segment. This section shows characteristics for 4 different cables which will be used later in this application note, and compares them to the standard.

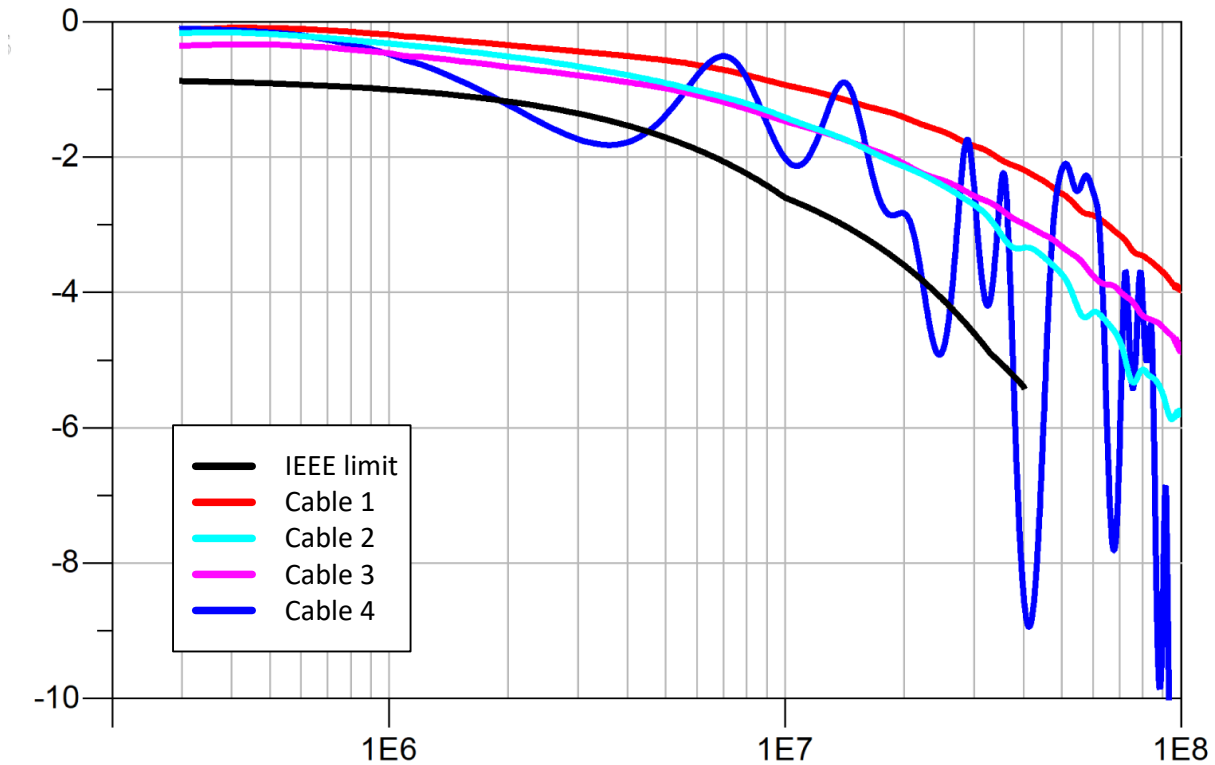
1. High quality single twisted pair with shield. Details available under request.
2. Single twisted pair cable. This particular cable is often used in Microchip demonstrations and trainings.
3. Single twisted pair from a standard CAT6 cable.
4. Low quality, single pair cable, no twists.

15 meter lengths were not available for all of the cables, so the characteristics were measured over 5 meters to create models which were then applied to a simulation tool. All simulations shown in this application note were generated using this same tool, and later sections will show that this tool corresponds very closely to actual measurements. Details of the simulation are available upon request.

Equation 4-1. Insertion Loss

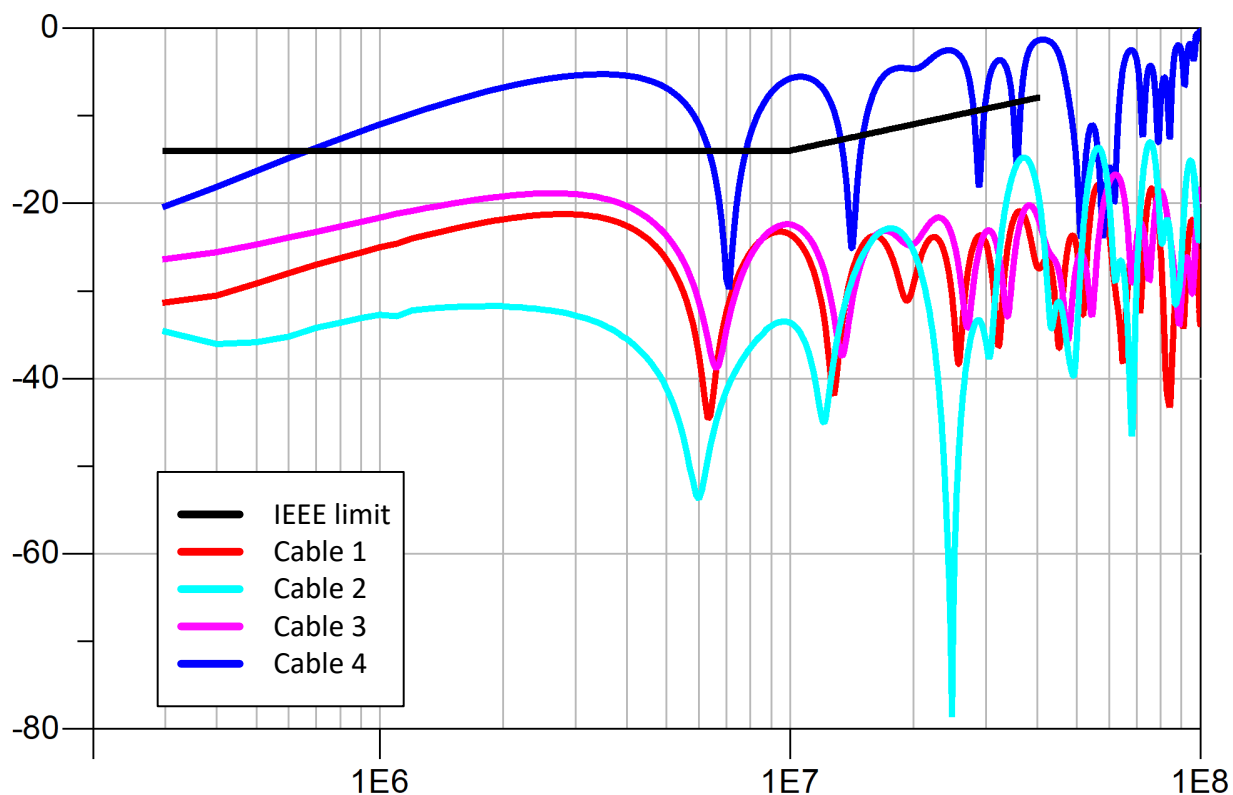
$$\text{Insertion Loss}(f) < \left\{ \begin{array}{ll} 1.0 + \frac{1.6(f-1)}{9} & 0.3\text{MHz} \leq f < 10\text{MHz} \\ 2.6 + \frac{2.3(f-10)}{23} & 10\text{MHz} \leq f < 33\text{MHz} \\ 4.9 + \frac{2.3(f-33)}{33} & 33\text{MHz} \leq f \leq 40\text{MHz} \end{array} \right\} \text{dB}$$

Insertion loss for the 4 cables above, along with the IEEE limit is shown in [Figure 4-1](#). All but cable 4 meet the requirement across all frequencies.

Figure 4-1. Insertion Loss for Different Cables**Equation 4-2.** Return Loss

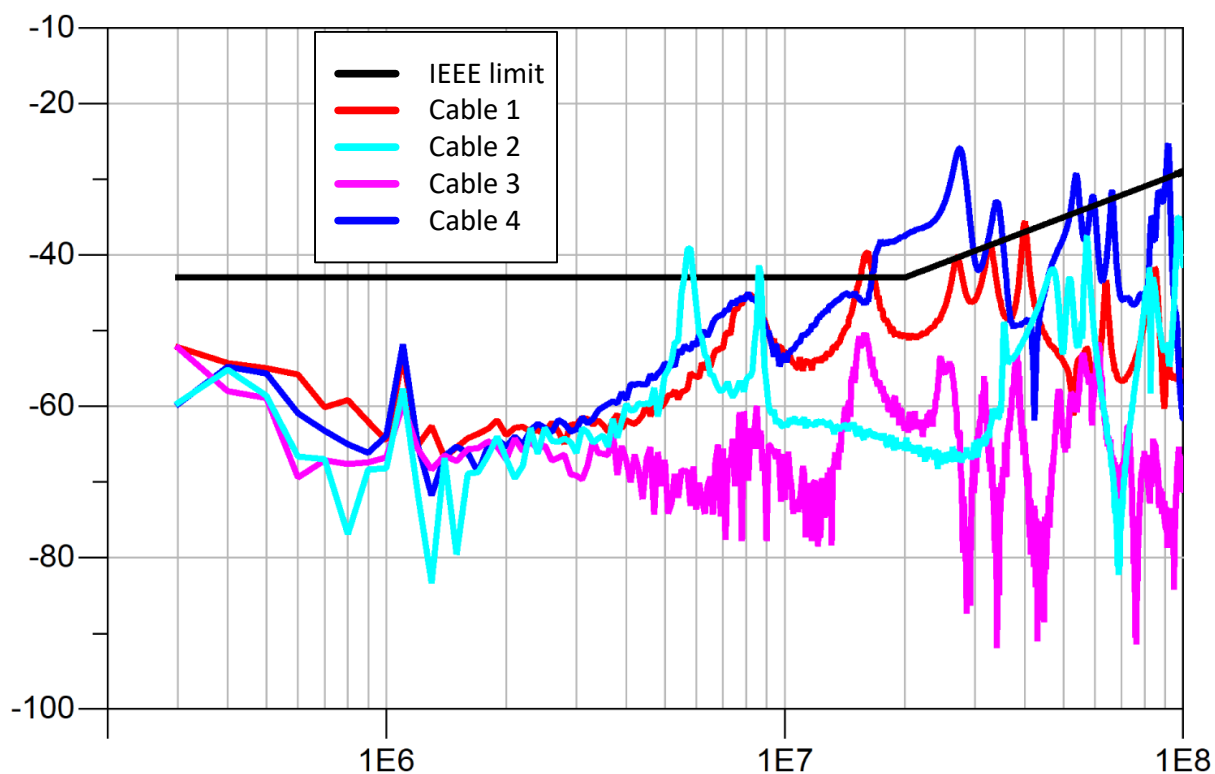
$$\text{Return loss}(f) > \begin{cases} 14 & 0.3\text{MHz} \leq f < 10\text{MHz} \\ 14 - 10\log_{10}\left(\frac{f}{10}\right) & 10\text{MHz} \leq f \leq 40\text{MHz} \end{cases} \text{dB}$$

The return loss for the 4 cables above is compared with the IEEE limit is shown in [Figure 4-2](#)

Figure 4-2. Return Loss for Different Cables**Equation 4-3.** Mode Conversion Loss

$$\text{Mode conversion loss}(f) > \begin{cases} 43 & 0.3\text{MHz} \leq f < 20\text{MHz} \\ 43 - 20\log_{10}\left(\frac{f}{20}\right) & 20\text{MHz} \leq f \leq 200\text{MHz} \end{cases} \text{dB}$$

Figure 4-3 shows the mode conversion loss and IEEE limits for the 4 cables. Note that only cable 3 meets this standard at all points.

Figure 4-3. Mode Conversion Loss for Different Cables

These are the characteristics that must be met at each node on the segment to ensure that compliant nodes will always work. Of course, compliant nodes may work on segments that do not meet these characteristics, but operation is not guaranteed.

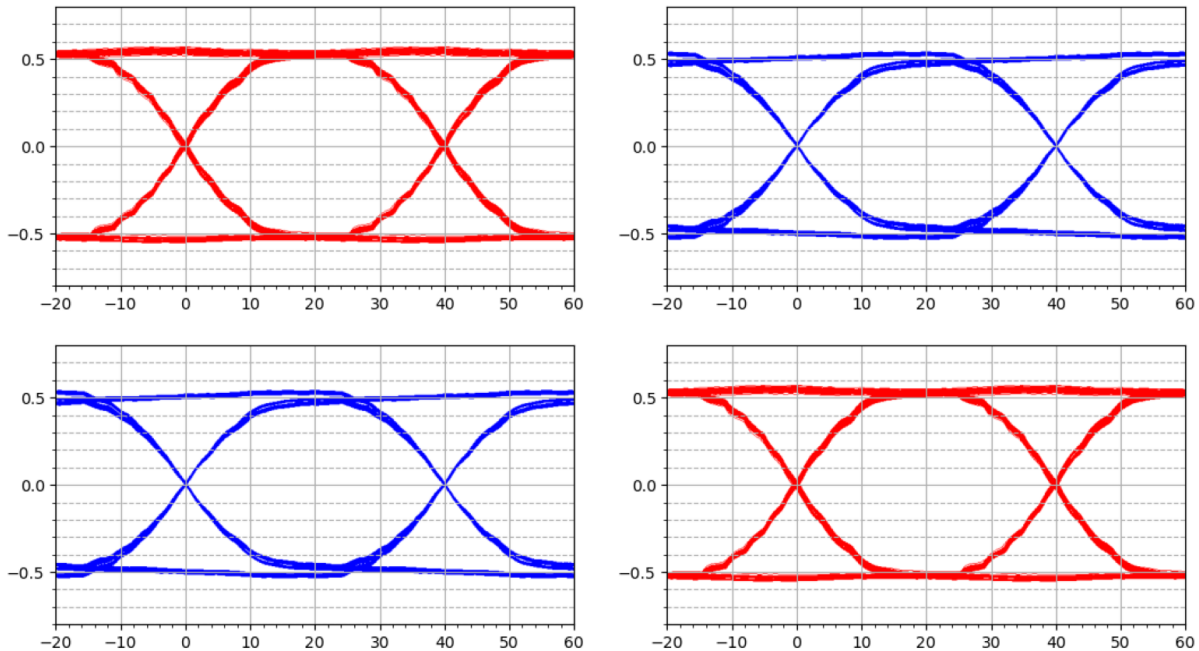
5. Starting Point: 5-Meter System with 2 Nodes

Understanding the effect of the inherent noise of a complex mixing segments begins with understanding a simple, 2 node example. The base line example for this application note will use 5 meters of cable between the nodes. The nodes are built using the min BIN configuration from the BIN Reference Design Application Note⁴. The cables join directly at the MDI of the node, so that there are no stubs. All of these systems work under laboratory conditions (ideal power supplies, room temperature, no alien noise).

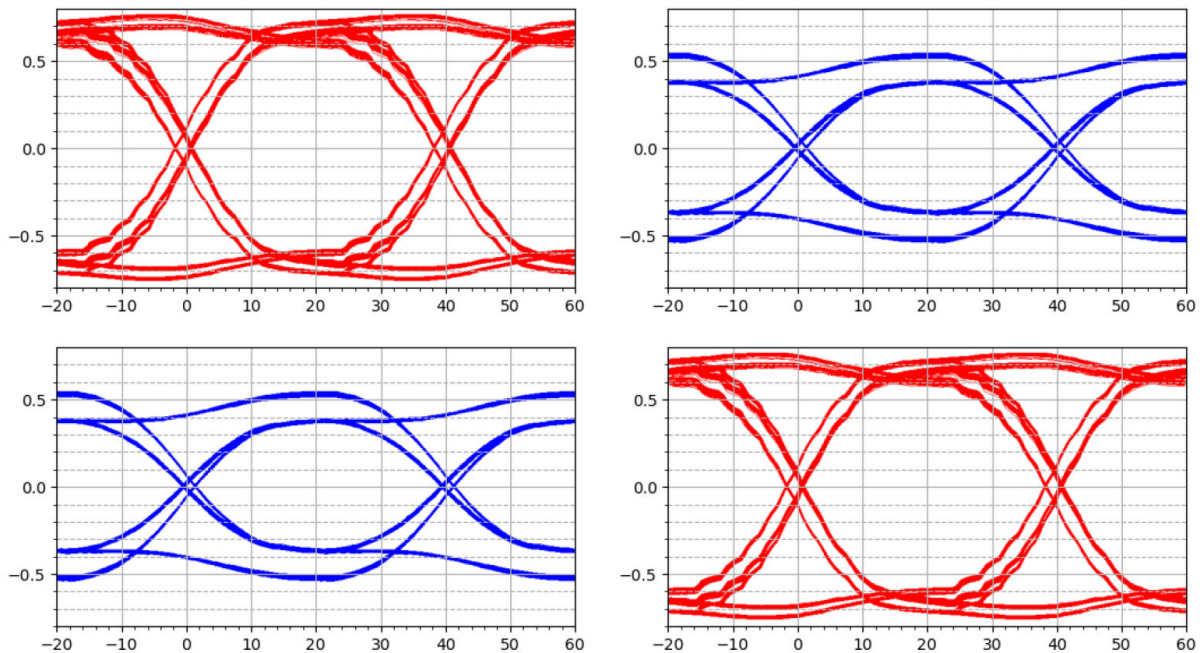
First, examine the eye diagrams of this simple system when using different cables. The eye patterns are different between each transmitter and each receiver. To study inherent noise, it is necessary to look at all of the eye patterns. The left column shows the eye when node 0 is transmitting, while the right column shows the eye when node 1 is transmitting. The first row of results is measured with the receiver at node0 and the 2nd row is measured with the receiver at node 1. The plots shown in red are made at the transmitter, and the plots in blue are at nodes that are receiving from the transmitter.

The results for cable 1 are shown in [Figure 5-1](#), while those for cable 4 are shown in [Figure 5-2](#).

Figure 5-1. Eye Diagrams for 5meters of Cable 1 with 2 Nodes

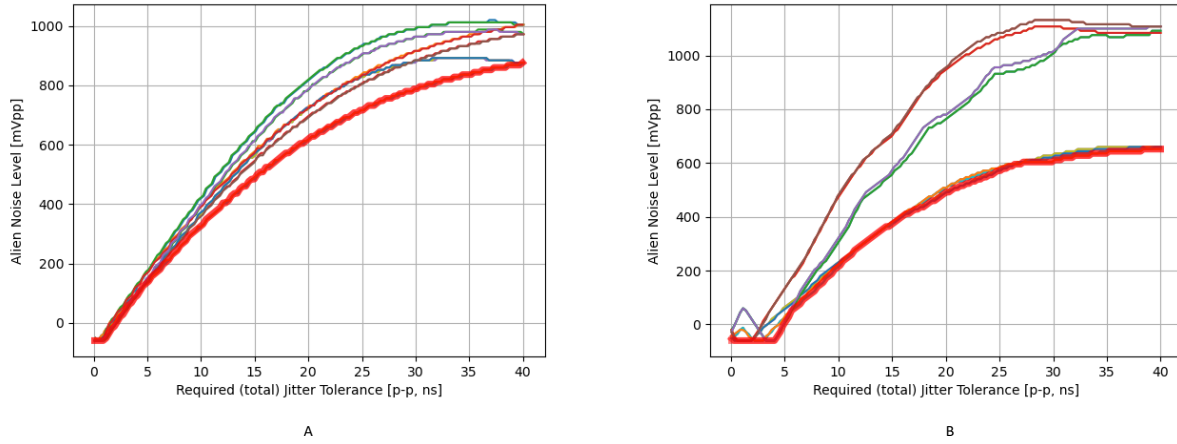


Cable 1 is a good quality cable. The signal at the receivers (blue plots) show a small amount of attenuation over the 5 meters, and no significant increase in signal jitter. Cables 2 and 3 produce similar results.

Figure 5-2. Eye Diagrams for 5 Meters of Cable 4 with 2 Nodes

Cable 4 is a poor quality cable. Even at the transmitters (the red plots), the eye patterns are showing signal distortion, with visible jitter, shown as multiple zero crossing and reduced noise margin as the maximum signal amplitude has significant variation. At the receivers, significant cable attenuation causes further signal distortion. Jitter is large, and the smaller eye opening reduces the noise margin. Under additional stress, such as more nodes or longer cable length, a system based on this cable will not have enough noise margin to properly function.

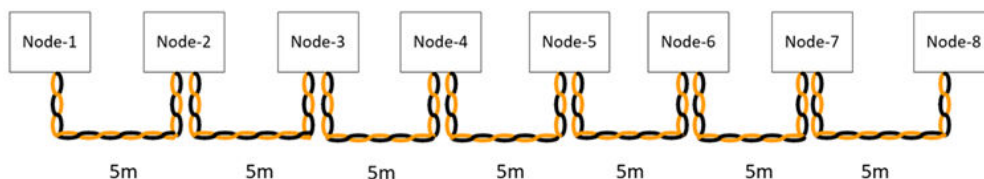
As the number of nodes in a system increases, the number of eye diagrams increases by the number of nodes squared. To make the analysis easier, the eye diagrams can be combined as shown in [Figure 5-3](#). This plot is created by dividing each diagram into 4 quadrants, rotating the quadrants and superimposing them on each other, and repeating the process for each of the $N \times N$ eye diagrams. The minimum height of the opening for any jitter value is used to create the red line, which is a plot of jitter tolerance versus maximum alien noise level. The 30mV offset of the eye patterns represents the 30mV differential input required by the receiver. The amplitude of the red curve above 0 is then the alien noise margin that remains after the worst case inherent noise. In the Figure, plot A shows the results for cable 1 and plot B shows the results for cable 4. In an automotive environment, a rule of thumb is to allow for at least 200mVpp of alien noise, so the poor quality cable 4 is already at its limit in this simple configuration for a jitter tolerance of less than 10ns.

Figure 5-3. Jitter Tolerance vs Noise Tolerance: 2 Nodes with 5 Meters of Cable

6. Longer Reach: 35-Meter System with 8 Nodes

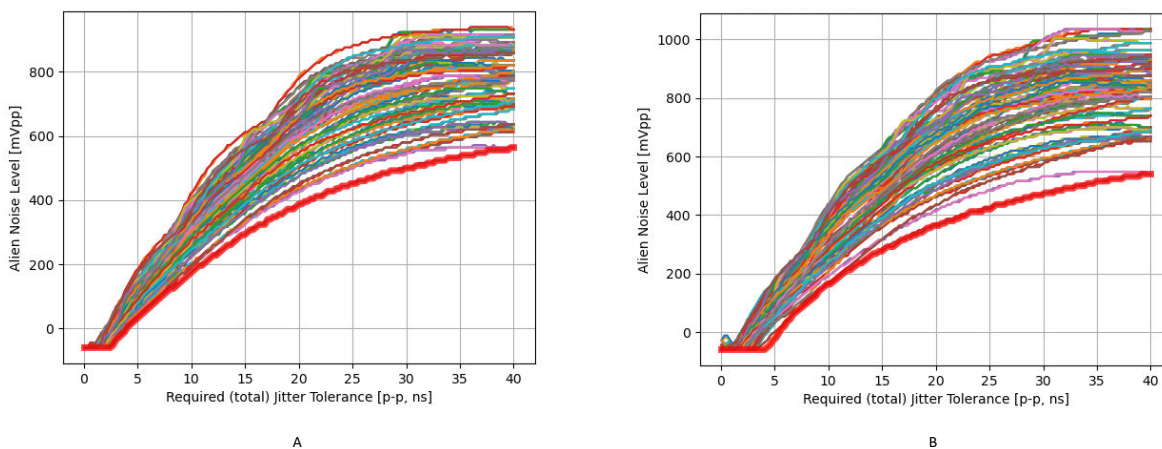
The same analysis can be used to consider the effect of adding more nodes and changing cable lengths. The next step used was to continue to use 5 meters of cable between each node, but add more nodes, for a total of 8 nodes as shown in [Figure 6-1](#).

Figure 6-1. 8 Node System with 5 Meters between Nodes



The resulting jitter tolerance vs noise tolerance plot is shown in [Figure 6-2](#), where plot A contains the measurements from cable 2 and plot B shows the results from cable 3. Both of these cables show good results, although, for this configuration, cable 2 has slightly better noise tolerance for systems that can tolerate peak to peak jitter above 12 ns. In addition to these measurements, communication between nodes of this system has been verified in the lab.

Figure 6-2. Jitter Tolerance vs Noise Tolerance: 8 Nodes with 35 Total Meters of Cable



7. Increasing Node Count, Extending Reach

Creating prototype systems and making measurements becomes more expensive and time consuming as the number of nodes increases. Further variations, like experimenting with different BINs, requires even more combinations and measurements. A small number of good reliable measurements enables the creation of reliable modes that can then be used in simulation tools to create realistic results and enable comparison of various network configurations. Simulations based on models that match the previous results created jitter vs noise tolerance plots, which can be used to identify configurations that have the desired noise tolerance at reasonable jitter margins. These simulations are used to determine working topologies.

The simulations were used to analyze two different scenarios using Microchip's LAN8670 with cable 2:

- High node density - how many nodes can be supported in a given condition, and how close can they be spaced.
- Maximum segment length - how long can a segment be, and how many (equally spaced) nodes can it support

High Node Density

When using a large number of nodes, an important parameter is how closely they can be to each other. The methods above were used to determine the minimum distance between nodes for a system, assuming a required noise tolerance of 200mV. If it is possible to avoid using a Common Mode Choke (CMC), minimum node spacing is 5 cm for up to 20 nodes, and 25 cm for up to 50 nodes, as shown in [Table 7-1](#). More nodes may be possible with lower noise tolerance.

Table 7-1. Minimum Distance between Nodes without CMC

Number of Nodes	Minimum Distance between Nodes	Total Length at Minimum Distance
8	5 cm	35 cm
10	5 cm	45 cm
20	5 cm	95 cm
40	25 cm	9.75 m
50	25 cm	12.25 m

A Common Mode Choke (CMC) causes the node spacing to increase, as shown in [Table 7-2](#).

Table 7-2. Minimum Distance between Nodes with CMC

Number of Nodes	Minimum Distance between Nodes	Total Length at Minimum Distance
8	20 cm	140 cm
10	40 cm	360 cm
20	60 cm	1140 cm
40	80 cm	3120 cm
50	90 cm	4410 cm

Maximum Segment Length

Maximum segment length, using the same configuration as above, is simpler. For 200mV noise tolerance, without a common mode choke, up to 50 nodes can support a total mixing segment length of 50 m. With a common choke, this drops to 45m.

If the end system can tolerate 100mV of noise, a 2 node system using cables 2 or 3 will function at a length of 100m. This has been shown working correctly under laboratory conditions.

8. Conclusions

With good quality cable and well designed nodes, it is possible to achieve up to 50 meters and up to 50 nodes using Microchips 10BASE-T1S product and have systems that function with high levels of noise immunity. In quiet environments, up to 100 meters have been demonstrated with 2 nodes.

To achieve reliable results, the mixing segment needs to be properly simulated. Simulation of the mixing segment begin with good models for the MDI combined with accurate cable models taken from laboratory measurements which are used to simulate the mixing segment. Simulation enables an understanding modeling of the system inherent noise, which makes it possible to determine if a configuration will work in a given noise environment.

9. References

1. IEEE Std 802.3cg™-2019, IEEE Standard for Ethernet, Amendment 5: Physical Layer Specifications and Management Parameters for 10 Mb/s Operation and Associated Power Delivery over a Single Balanced Pair of Conductors. standards.ieee.org/standard/802_3cg-2019.html
2. LAN8670/1/2 10BASE-T1S Ethernet PHY Transceiver Data Sheet, DS60001573, www.microchip.com/LAN8670
3. LAN8650/1 10BASE-T1S MAC-PHY Ethernet Controller with SPI Datasheet, DS60001734 www.microchip.com/LAN8650
4. LAN86xx Bus Interface Network (BIN) Reference Design Application Note, www.microchip.com/DS60001718

10. Revision History

Table 10-1.

Revision Level & Date	Section/Figure/Entry	Changes
DS60001829B (Oct-2023)	Title	Added AN-1829 to title. No change in content.
DS60001829A (Oct-2023)	All	Initial revision

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